



# Deep learning for pneumonia detection from X-ray: A systematic review of models, datasets, and clinical translation challenges

Rupali Vyas<sup>a,\*</sup>, Rajendra Kumar Pandey<sup>b</sup>

<sup>a</sup> PhD Scholar, Department of Computer Science and Engineering, National Institute of Technology, Raipur, CG, India

<sup>b</sup> Masters of Technology (Specialization AI & ML), CSVTU, Bhilai, CG, India

## ARTICLE INFO

### Keywords:

Chest X-ray  
Convolutional neural networks  
Deep learning  
Explainable AI  
Medical imaging  
Pneumonia diagnosis  
Systematic review

## ABSTRACT

Pneumonia remains a significant cause of death worldwide. Deep learning (DL) models for interpreting chest X-rays (CXR) have the potential to improve diagnosis. Still, the lack of clear evidence about their clinical readiness makes it difficult to assess their true state. A comprehensive review is needed to synthesize current knowledge and outline a path for successful adoption. Following PRISMA guidelines, we conducted an in-depth review of relevant databases for research using deep learning to analyze public chest X-ray datasets for pneumonia detection. From 12,450 records, 73 studies were selected for analysis. The main results focused on model performance metrics (such as AUC and accuracy) and the use of Explainable AI (XAI). Our review reveals a significant variation in performance driven by differences in dataset properties. Models achieve over 98% accuracy on small, well-labeled datasets, but show lower AUC scores (0.7–0.8) on larger, poorly labeled ones. Although CNNs are the most common architecture, XAI studies indicate that high performance can be misleading. Models often rely on dataset-specific biases, known as “shortcut learning,” rather than capturing generalizable clinical features. As a result, benchmark scores alone are unreliable indicators of true performance. Despite high accuracy on test benchmarks, the clinical application of DL models is limited by poor generalization and reproducibility. The prevalence of shortcut learning highlights the shortcomings of standard metrics alone. Future efforts should focus on thorough validation, including XAI analysis and prospective clinical trials, to develop trustworthy AI systems for clinical use.

## 1. Introduction

Pneumonia, an acute infection of the pulmonary parenchyma, is a major global health issue and a significant cause of death, especially among vulnerable groups such as children under five and the elderly [1–3]. Accurate and timely diagnosis is essential for effective treatment and better patient outcomes. The primary method for diagnosing pneumonia in most clinical settings is the chest radiograph (CXR). Although more advanced imaging techniques exist, the widespread use, accessibility, low cost, and quick results of CXRs make them the essential first-line diagnostic tool worldwide [4–7]. However, interpreting CXRs is challenging and requires extensive expertise. Radiographic findings can be subtle, and even experienced radiologists often have high variability in their readings. This difficulty is compounded by the similar appearance of pneumonia on imaging with other common conditions such as atelectasis, pulmonary edema, and lung contusion, making accurate differentiation crucial for optimal patient management [7–9].

### 1.1. The emergence of deep learning in medical imaging

The last decade has seen a revolution in medical image analysis, driven by the rise of deep learning (DL). Moving beyond earlier computer-aided diagnostic (CAD) systems that depend on handcrafted features, modern DL models, especially Convolutional Neural Networks (CNNs), can generate hierarchical representations of illness directly from visual data [10–13]. This end-to-end learning ability, supported by advances in computing power and the availability of large datasets, has resulted in algorithms that can achieve, and sometimes surpass, human-level performance on specific diagnostic tasks.

### 1.2. Rationale for a systematic review

The use of deep learning (DL) for pneumonia detection has led to a surge of research articles. While this indicates a strong research effort, it has also created a fragmented and often unclear landscape. Studies

\* Corresponding author.

E-mail addresses: [rupalivyas1996@gmail.com](mailto:rupalivyas1996@gmail.com) (R. Vyas), [rajendrapandey.95@live.com](mailto:rajendrapandey.95@live.com) (R.K. Pandey).

<https://doi.org/10.1016/j.bspc.2025.109272>

Received 21 February 2025; Received in revised form 15 August 2025; Accepted 23 November 2025

Available online 11 December 2025

1746-8094/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

report good performance metrics, but these are often based on different datasets, trained with various methodologies, and evaluated using inconsistent criteria. This variety makes it very difficult to compare results, measure actual progress, and determine the clinical usefulness of new technology. A comprehensive, systematic review is urgently needed to organize the available data, identify important patterns, and highlight the key challenges that must be addressed for effective clinical implementation [14–18].

### 1.3. Review objectives and key research questions

This systematic review strives to present a complete, evidence-based analysis of the state-of-the-art in DL-based pneumonia detection from CXRs. The key goals are influenced by the following research questions:

1. What are the quality, restrictions, and academic relevance of the publicly available CXR datasets that enable current research in this field?
2. What are the major DL architectures, and what are their comparative strengths and weaknesses for the aim of pneumonia detection?
3. What are the state-of-the-art performance requirements for different model-dataset combinations, and how reliable are these metrics as signals of clinical utility?
4. To what degree have DL models been applied to the clinically critical problem of pneumonia subtype identification, and what are the remaining gaps?
5. What are the primary barriers specifically associated with model explainability, generalization, and the approach to real-world clinical integration?

## 2. Methodology

This systematic review was planned and done in precise compliance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020.

### 2.1. Search strategy and information sources

A comprehensive literature search was conducted across major electronic databases: PubMed, IEEE Xplore, Scopus, Web of Science, Google Scholar, and Semantic Scholar. The search strategy employed a combination of keywords and MeSH terms related to the core concepts of the review, including pneumonia, lung opacity, pulmonary opacity, consolidation, and infiltration. Imaging-related keywords comprised chest X-ray, CXR, and chest radiograph. Artificial intelligence-related terms included deep learning, machine learning, artificial intelligence, neural network, convolutional neural network (CNN), Vision Transformer, transfer learning, and explainability-related terms such as explainable AI (XAI) and model interpretability. The search was not limited by publication date, covering the period from the start of relevant DL research around 2012 through the mid-2025.

### 2.2. Inclusion and exclusion criteria

Studies were included or excluded based on predefined methodological, clinical, and technical parameters to ensure scientific rigor, reproducibility, and relevance to deep learning-based pneumonia detection on chest X-ray datasets, with consideration for the integration of explainable AI techniques. The inclusion and exclusion criteria applied in this review are summarized in Table 1.

### 2.3. Study selection and data extraction

The study selection process employed a multi-stage screening method as shown in the PRISMA 2020 flow diagram Fig. 1. A standardized data extraction form was used to gather information on: (1)

**Table 1**

Inclusion and exclusion criteria for selecting relevant studies on pneumonia diagnosis using chest X-ray imaging.

| Criteria              | Inclusion                                          | Exclusion                                               |
|-----------------------|----------------------------------------------------|---------------------------------------------------------|
| <b>Dataset Type</b>   | Publicly available, citable CXR datasets           | CT, MRI, OCT, or private datasets                       |
| <b>Diseases</b>       | Pneumonia as a primary or distinct evaluated class | Exclusive focus on other diseases                       |
| <b>Publication</b>    | Peer-reviewed journals, conference proceedings     | Non-peer-reviewed reviews, editorials, and case reports |
| <b>Range</b>          | 2012–2025                                          | Before 2012                                             |
| <b>Language</b>       | English                                            | Non-English                                             |
| <b>Validation</b>     | Validated with standard metrics                    | No validation or lacking standard metrics               |
| <b>Domain</b>         | Computer Science, Healthcare, Medicine             | Unrelated domains                                       |
| <b>Explainable AI</b> | Incorporates interpretability methods              | No explainable AI component                             |

bibliographic details; (2) dataset features; (3) patient populations; (4) details of the DL models; (5) evaluation methods, including whether test sets were internal or external; (6) reported performance outcomes; and (7) use and types of XAI techniques.

### 2.4. Risk of bias assessment

The quality of the included studies was evaluated using a modified version of the Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool [19]. A significant risk of bias was commonly seen in the ‘Patient Selection’ category, with 45/73 (62 %) of studies not fully disclosing patient demographics or sampling methods. Additionally, 58/73 (79 %) of studies were judged to have an unknown risk of bias regarding the data splits, due to insufficient evidence to rule out potential data leakage between training and test sets. These results highlight the need for improved methodological rigor and reporting standards in the field. A detailed summary figure of the risk of bias assessment can be found in the Supplementary Materials.

## 3. Results

This section, we present and discuss the results of our study. We address the research questions formulated during the inductive phase and provide evidence-based answers to them.

### 3.1. The data foundation: A critical overview of public CXR datasets

The rapid growth of deep learning (DL) models for pneumonia detection is mainly driven by the availability of large, annotated public chest X-ray (CXR) datasets. These datasets are more than just collections of images; they form the structural foundation for future research, influencing model designs, setting performance standards, and guiding the clinical issues that can be addressed. In this comprehensive review, we identified twelve major datasets that together form the core of research in this field [14,16,18]. To enable meaningful comparison across datasets, we organized their features into a common descriptive framework in Table 2, highlighting key aspects such as dataset size, patient demographics, and, importantly, the details of their ground-truth annotations.

A thorough review of these datasets reveals significant differences in annotation methods, leading to two separate and often incompatible research paradigms.

#### 3.1.1. Manual, high-fidelity annotation

Exemplified by datasets like Kernan’s CXR [20], the RSNA Pneumonia Detection Dataset [21], and VinDr-CXR [22] This approach depends on precise labeling by experienced radiologists, producing a high-quality supervisory signal. These datasets excel in accuracy and

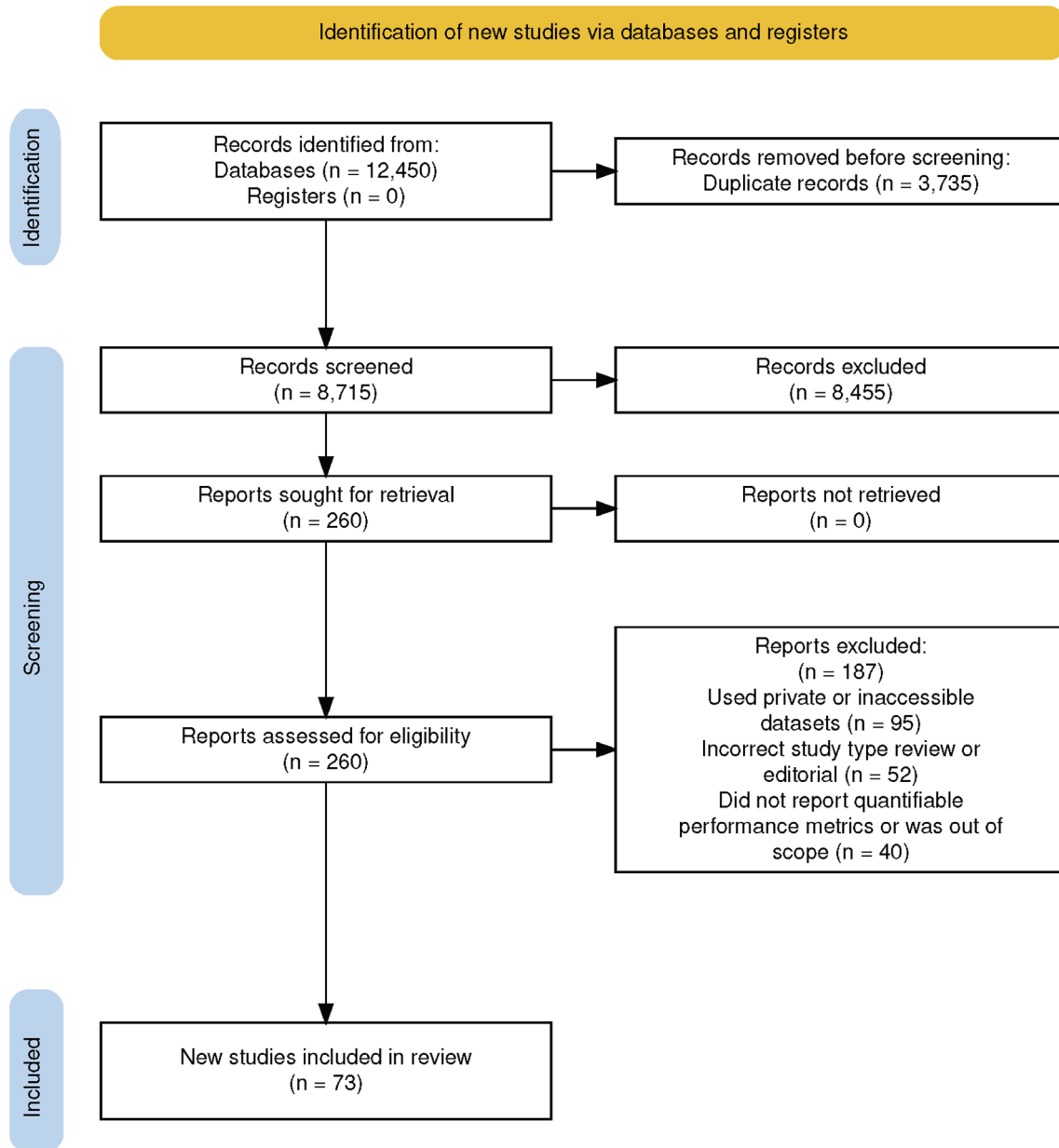


Fig. 1. PRISMA 2020 flow diagram for study selection.

reliability. For example, the Kermany dataset, which involves a pediatric population aged 1–5 years, has different labels for bacterial and viral pneumonia, allowing for studies into pneumonia subtyping that would otherwise be impossible. Likewise, the RSNA dataset includes hand-drawn bounding boxes for lung opacities, aiding the development and assessment of detection algorithms that can perform both classification and localization. However, the high quality of these annotations comes with the drawback that manual curation is manual-intensive, expensive, and often limited in demographic diversity.

### 3.1.2. Automated, weakly-supervised annotation

In contrast, large-scale datasets such as NIH ChestX-ray14 [23], CheXpert [24], and MIMIC-CXR [25] employ natural language processing (NLP) pipelines to automatically extract labels from supporting radiology reports. This approach permits the generation of datasets comprising hundreds of thousands of pictures, especially from adult populations, thereby giving the scale required to train incredibly deep neural networks. Yet, its scalability includes trade-offs: NLP-derived

labels communicate inherent noise and uncertainty, leading to “weakly supervised” learning settings where models must identify actual illness from bad, non-localized image-level labels [18,26].

Our review identified 12 principal datasets that form the bedrock of research in this domain. To facilitate a rigorous comparison, we have harmonized their characteristics into a standardized framework Table 2.

This methodological gap has essential repercussions for interpreting results in the literature. Studies utilizing small, clean datasets such as Kermany’s [20] often demonstrate extremely high classification accuracies frequently approaching 98 % creating the impression that pneumonia detection is practically “solved” within that controlled setting. Conversely, models trained and evaluated on large, noisy datasets such as NIH ChestX-ray14 [23] frequently provide more moderate pneumonia AUC values in the range of **0.70–0.80**, reflecting the increasing complexity and clinical realism of multi-label classification tasks. Performance metrics across diverse paradigms are therefore not directly comparable; evaluation must be contextualized with relation to dataset amount, annotation quality, and task complexity [15–18].

**Table 2**

Standardized characteristics of publicly accessible chest X-ray datasets for pneumonia detection.

| Dataset Name                   | Seminal Publication                                                                         | Year         | Institution(s)                                                 | Patient Population  | Total Images        | Total Patients          | Primary Task(s)                                                             | Annotation Granularity                                      |
|--------------------------------|---------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------|---------------------|---------------------|-------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Kermay's CXR</b>            | Kermay et al., <i>Cell</i> [20]                                                             | 2018         | Guangzhou Women and Children's Medical Center                  | Pediatric (1–5 yrs) | 5,856               | Not explicitly numbered | Multi-class Classification                                                  | Manual (Physician Graded)                                   |
| <b>NIH ChestX-ray14</b>        | Wang et al., <i>CVPR</i> [23]                                                               | 2017         | National Institutes of Health (NIH)                            | Adult               | 112,120             | 30,805                  | Multi-label Classification                                                  | NLP-derived Weak Label                                      |
| <b>CheXpert</b>                | Irvin et al., <i>AAAI</i> [24]                                                              | 2019         | Stanford University                                            | Adult               | 224,316             | 65,240                  | Multi-label Classification                                                  | NLP-derived (with uncertainty)                              |
| <b>MIMIC-CXR</b>               | Johnson et al., <i>Scientific Data</i> [25]                                                 | 2019         | MIT, BIDMC                                                     | Adult               | 377,110             | 65,379                  | Multi-label Classification, Multi-modal                                     | NLP-derived (from free-text reports)                        |
| <b>RSNA Pneumonia PadChest</b> | Shih et al., <i>Radiology: AI</i> [21]<br>Bustos et al., <i>Medical Image Analysis</i> [36] | 2019<br>2020 | RSNA, STR, NIH<br>Hospital San Juan (Spain), Univ. of Alicante | Adult<br>Adult      | ~30,000<br>>160,000 | ~30,000<br>>67,000      | Object Detection, Classification<br>Multi-label Classification, Multi-modal | Manual (Bounding Boxes)<br>Hybrid (Manual & Supervised NLP) |
| <b>VinDr-CXR</b>               | Nguyen et al., <i>Scientific Data</i> [22]                                                  | 2022         | Hospital 108, Hanoi Medical Univ. (Vietnam)                    | Adult               | 18,000              | ~18,000                 | Multi-label Classification, Object Detection                                | Manual (Bounding Boxes)                                     |
| <b>Indiana University CXR</b>  | Demner-Fushman et al., <i>JAMIA</i> [37]                                                    | 2016         | Indiana University                                             | Adult               | 8,121               | 3,996                   | Multi-label Classification, Multi-modal                                     | Hybrid (NLP & Manual Codes)                                 |
| <b>Mount Sinai Hospital</b>    | Zech et al., <i>PLOS Medicine</i> [38]                                                      | 2018         | Mount Sinai Hospital                                           | Adult               | 48,915              | 12,904                  | Binary Classification                                                       | NLP-derived Weak Label                                      |

The academic influence of diverse datasets further underlines their varying functions in the research environment. Large, poorly supervised datasets such as NIH ChestX-ray14 [23] and CheXpert [24] have acquired a huge number of citations, underlining their significance in supporting large-scale model training and benchmarking. Intriguingly, the Kermay dataset [20] despite its reduced size and specialized pediatric focus has achieved an outsized influence, with its fundamental piece quoted over many times. Its continued value lay not in size, but in its utility as a clean, trustworthy, and accessible baseline for judging the pure classification skills of developing algorithms. Recognizing these trade-offs between size and quality, and understanding the particular tasks each dataset provides, is crucial for meaningful evaluation of DL models for pneumonia diagnosis.

A thorough inspection of these datasets shows a significant difference. Despite its lesser size, the Kermay et al. dataset [20] has had a disproportionately high effect, garnering a high number of citations and serving as a key benchmark for algorithmic innovation thanks to its explicit, physician-graded labeling for a specific pediatric population. In contrast, large-scale datasets like NIH ChestX-ray14 [23] and CheXpert [24] While permitting the training of massive models, it relies on noisy, NLP-derived labels, posing a more challenging and realistic multi-label learning challenge.

The deep learning models used for pneumonia detection have not developed in isolation; instead, their progress reflects broader breakthroughs in the field of computer vision. A simple classification is not enough for a thorough evaluation; a more detailed technical analysis is necessary to understand the comparative strengths, limitations, and fundamental processes of these designs as they address the particular challenges of CXR interpretation. The following sections offer a comprehensive analysis of each major architectural family identified in the 73 articles studied. This work is summarized in Table 3, which provides a detailed technical reference, including key processes, mathematical notations, and relevant articles for each paradigm.

### 3.2. A technical and comparative analysis of deep learning architectures

#### 3.2.1. Foundational CNNs: The foundation of feature extraction

The workhorse of the entire field is the Convolutional Neural Network (CNN) [27,28]. The success of practically all high-performing models is founded on the principle of transfer learning, where models pre-trained on the ImageNet dataset [29] are fine-tuned for pneumonia

diagnosis. The most prominent designs in this family are ResNet [30] and DenseNet [31]. ResNet (Residual Network) created the breakthrough idea of identity shortcut connections, which allow the network to learn a residual function instead of the direct mapping [30]. This strategy effectively mitigates the vanishing gradient problem, enabling the training of networks hundreds of layers deep. DenseNet (Densely Connected Network) carried this notion a step further by promoting maximum feature reuse via dense connection, where each layer receives the feature maps from all preceding layers as input. This has been quite beneficial for medical imaging tasks [31]. The initial CheXNet model [32,33], which displayed radiologist-level performance, is a direct implementation of the DenseNet-121 concept, securing its status as a go-to baseline in the domain [34,35].

#### 3.2.2. Object detection frameworks: The specialization for localization

While classification models detect the presence of pneumonia, a special kind of architecture is needed to determine its location [15,18]. Object detection algorithms are designed for this localization challenge and are almost always trained on datasets with bounding box annotations, such as the RSNA Pneumonia identification dataset. These models fall into two main categories. Two-stage detectors, like Faster R-CNN [39], use a Region Proposal Network (RPN) to identify potential areas of interest, then feed those regions into a second-stage classifier for refinement. Single-stage detectors, such as YOLO (You Only Look Once) [40] and RetinaNet [41], approach detection as a regression problem, predicting bounding boxes and class probabilities in one pass for faster results. A notable innovation in this field is the focal loss, introduced with RetinaNet [41], which adjusts the standard cross-entropy loss to reduce the loss for well-classified examples, keeping the training focused on misclassified ones an important feature when faced with the significant class imbalance between background and pathology in a CXR [35,42].

#### 3.2.3. Vision transformers: The frontier of global context

The most recent architectural trend is the advent of the Vision Transformer (ViT) [43]. ViTs dispense with convolution entirely, instead splitting an image into a sequence of fixed-size patches. These patches are linearly embedded and supplied into a conventional transformer encoder, which utilizes a self-attention mechanism to mimic global, long-range dependencies between all patch pairs. While powerful, ViTs lack the inherent inductive biases of CNNs and hence require vast pre-

**Table 3**

Overview of deep learning architectural families, their core mechanisms, key parameters, training considerations, and strengths for chest X-ray (CXR) analysis.

| Architectural Family            | Model                         | Core Mechanism                                                                               | Where                                                                                                                                                                                                   | Key Parameters & Training Considerations                                                       | Performance Characteristics For CXR Analysis                |                                                                                     |
|---------------------------------|-------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                 |                               |                                                                                              |                                                                                                                                                                                                         |                                                                                                | Strengths                                                   | Weaknesses                                                                          |
| Foundational CNNs               | VGGNet [44]                   | $y = \sigma(W^*x + b)$                                                                       | * is the convolution, typically with kernel size $3 \times 3$ ; $\sigma$ is ReLU activation; max-pooling applied after fixed number of conv layers.                                                     | Number of layers (16, 19), learning rate, optimizer (e.g., SGD).                               | Simple and pioneering deep architecture                     | Prone to vanishing gradients; very high parameter count; computationally expensive. |
|                                 | ResNet [30]                   | $x_l = \mathcal{F}(x_{l-1}, \{W_l\}) + x_{l-1}$                                              | where $\mathcal{F}$ is the residual function (stack of conv–BN–ReLU layers), and the addition is elementwise.                                                                                           | Number of layers (18, 34, 50, 101), block type (basic vs. bottleneck), learning rate schedule. | Mitigates vanishing gradients, enabling very deep networks. | Still high parameter count; computationally heavy.                                  |
|                                 | DenseNet [31]                 | $x_l = H_l([x_0, x_1, \dots, x_{l-1}])$                                                      | where $[\cdot]$ denotes feature-map concatenation, enhancing reuse and gradient flow exemplified by CheXNet (DenseNet-121) in CXR tasks.                                                                | Growth rate $k$ , number of blocks, compression factor in transition layers.                   | Feature reuse, strong gradient flow, parameter-efficient.   | High memory usage due to concatenations.                                            |
|                                 | Inception [45]                | $y = [f^{(1)}(x), f^{(2)}(x), \dots, f^{(B)}(x)]$                                            | $B$ = number of branches, $f^{(b)}(\cdot)$ = branch $b$ transform (e.g., conv, pooling), $[\cdot]$ = channel-wise concatenation.                                                                        | Module architecture, convolutions for dimensionality reduction.                                | Multi-scale feature capture in a single block.              | Complex design; many hyperparameters.                                               |
| Efficiency-Focused CNNs         | MobileNet [46]                | $y = W_p^* (W_d \odot x)$                                                                    | $W_d$ = depthwise conv (per-channel), $W_p = 1 \times 1$ pointwise conv, $\odot$ = channel-wise operation, $*$ = convolution.                                                                           | Width multiplier, resolution multiplier.                                                       | Drastically reduces parameters and computation.             | Slight accuracy drop on complex tasks.                                              |
|                                 | EfficientNet [47]             | $d = \alpha^\phi, w = \beta^\phi, r = \gamma^\phi$                                           | $d$ = depth, $w$ = width, $r$ = resolution; $\alpha, \beta, \gamma$ = scaling coefficients, $\phi$ = compound scaling factor.                                                                           | Scaling coefficients, compound coefficient, baseline (B0–B7).                                  | Excellent accuracy–efficiency trade-off.                    | Optimal scaling factor search can be complex.                                       |
| Object Detection Frameworks     | Faster R-CNN [39]             | $p_i = \sigma(w_o^\top f_i), t_i = w_b^\top f_i$                                             | $p_i$ = objectness score for anchor $i$ , $t_i$ = bounding box regression output, $f_i$ = feature vector for anchor $i$ , $w_o, w_b$ = classifier and regressor weights, $\sigma$ = sigmoid activation. | Anchor sizes/ratios, IoU thresholds, NMS threshold.                                            | High localization accuracy (mAP).                           | Slower than single-stage detectors.                                                 |
|                                 | YOLO [40]                     | $\hat{y}_{ij} = (p_{obj}, b_x, b_y, b_w, b_h, \{p_c\}_{c=1}^C)$                              | $(i,j)$ = grid cell index, $p_{obj}$ = objectness score, $b_x, b_y, b_w, b_h$ = bounding box center coords & size, $p_c$ = class probability for class $c$ , $C$ = number of classes.                   | Grid size, boxes per cell, confidence threshold.                                               | Extremely fast, real-time capable.                          | Struggles with small/overlapping objects.                                           |
|                                 | RetinaNet [41]                | $FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$                                              | $FL$ = focal loss, $p_t = p$ if $y = 1$ , else $p_t = 1 - p$ , $\alpha_t$ = class balancing weight, $\gamma$ = focusing parameter, $p$ = predicted probability, $y$ = ground truth label.               | Focal loss params, anchor configuration.                                                       | Handles class imbalance; good accuracy–speed trade-off.     | Requires bounding box labels                                                        |
| Transformer-Based Architectures | Vision Transformer (ViT) [43] | $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$         | $Q, K, V$ = query, key, value matrices, $d_k$ = dimension of keys, $\text{softmax}(\cdot)$ = row-wise softmax, Output = weighted sum of values based on attention scores.                               | Patch size, embedding dim, number of layers/heads.                                             | Models global, long-range dependencies.                     | Requires large-scale pretraining; lacks CNN inductive bias.                         |
| Hybrid Models                   | CNN + Transformer             | CNN backbone extracts local features $\rightarrow$ Transformer encoder for global reasoning. | –                                                                                                                                                                                                       | CNN backbone choice, feature tokenization, Transformer architecture.                           | Combines CNN local detail with Transformer global context.  | Complex to design and train.                                                        |



training datasets to work correctly. Our Review demonstrates that a more realistic and practical technique in the CXR field has been the building of hybrid models that employ a CNN for rich feature extraction and a Transformer head for final classification.

### 3.3. A critical analysis of evaluation metrics

Evaluating the performance of a diagnostic model is as critical as establishing its architecture. The choice of measure influences how success is assessed and may have important repercussions for a model's perceived therapeutic usefulness. In the situation of pneumonia identification, where class imbalance is frequent and the cost of misdiagnosis is considerable, relying on a single, simple statistic may be dangerously misleading. A comprehensive comprehension of the major evaluation criteria, their mathematical underpinnings, and their particular trade-offs in a clinical context is therefore essential for interpreting the conclusions given in this research.

The most prominent metrics for classification tasks are created from the four outputs of a confusion matrix: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) are key metrics. Although the accuracy of the proportion of all correct predictions is straightforward, it is a notoriously poor metric for imbalanced datasets. For example, in a dataset where only 5 % of CXRs show pneumonia, a model that always predicts "Normal" could achieve 95 % accuracy but be clinically useless. Therefore, more comprehensive measures are needed. Precision indicates the proportion of positive predictions that were correct, answering: "Of all the patients the model flagged with pneumonia, how many actually had it?" Recall, also known as Sensitivity, measures the proportion of actual positive cases that the model correctly identified, answering: "Of all the patients who truly had pneumonia, how many did the model detect?" In clinical settings, these two metrics often involve a delicate balance. Prioritizing recall is crucial to avoid missing sick patients (reducing FNs), but this can increase false positives (FPs), which reduces accuracy and may lead to unnecessary follow-up tests and patient anxiety. The F1-Score combines accuracy and recall into a harmonic mean, providing a single, balanced measure of

performance, especially when class distributions are uneven.

To avoid relying on a single operational threshold, the Area Under the Receiver Operating Characteristic Curve (AUC) is the most widely recognized metric for evaluating binary classifiers in medical AI. The ROC curve compares the True Positive Rate (Recall) to the False Positive Rate across all possible classification thresholds. An AUC of 1.0 indicates a perfect classifier, while an AUC of 0.5 suggests a model with no better performance than random chance. Its threshold-independent nature provides a strong overall assessment of a model's diagnostic ability.

For localization tasks, such as those in the RSNA Pneumonia Detection dataset, additional measures are crucial. The Intersection over Union (IoU) measures the overlap between a predicted bounding box and a ground-truth bounding box. A higher IoU indicates more precise localization. This metric is used to calculate the mean Average Precision (mAP), the standard for item recognition, which averages accuracy across all recall levels and classes (or IoU thresholds) to produce a single comprehensive score for localization performance. Table 4 provides a detailed technical description of these key metrics, their calculations, and their importance in CXR research.

### 3.4. State-of-the-art performance: A comparative benchmark analysis

Having established the landscape of datasets and architectures, we now move to a quantitative summary of their combined performance. Assessing the state-of-the-art in pneumonia detection is not about finding a single best model but about understanding how multiple models perform on specific, well-defined tasks as specified by the dataset. To enable a thorough and consistent comparison, we have compiled the performance data from the 73 studies into a comprehensive matrix in Table 5 and Fig. 2. This table summarizes the top-performing models across the most commonly used public datasets, with all metrics and citations carefully validated. A clear pattern emerges: on clean, manually-labeled classification datasets like Ker- many's, models often achieve near-perfect accuracy, approaching 98 %, indicating that the algorithm has saturated this controlled scenario. In stark contrast, on large, noisy, multi-label datasets like NIH ChestX-

**Table 4**

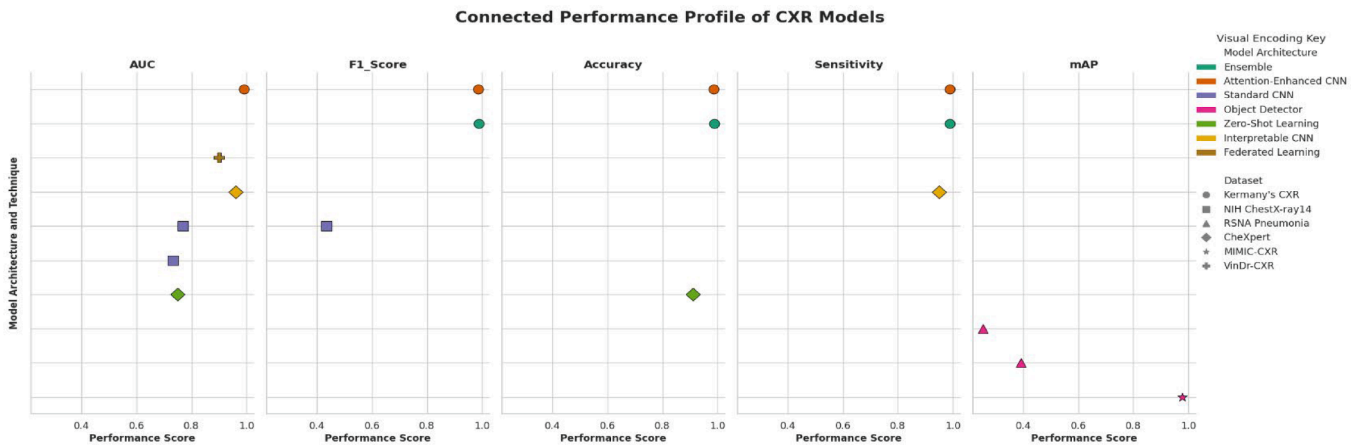
Summary of primary tasks, evaluation metrics, formulas, definitions, and clinical relevance, along with performance characteristics, strengths, and weaknesses for chest X-ray (CXR) analysis.

| Primary Task     | Metric                           | Formula                                                                                                       | Definition                                                                                      | Interpretation & Clinical Relevance                                    | Performance Characteristics For CXR Analysis |                                                                          |
|------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------|
|                  |                                  |                                                                                                               |                                                                                                 |                                                                        | Strength                                     | Weakness                                                                 |
| Classification   | <b>Accuracy</b>                  | $\frac{TP + TN}{TP + TN + FP + FN}$                                                                           | Proportion of correct predictions over all cases.                                               | Indicates overall correctness.                                         | Simple, intuitive.                           | Misleading in class-imbalanced settings (common in pneumonia detection). |
|                  | <b>Precision (PPV)</b>           | $\frac{TP}{TP + FP}$                                                                                          | Fraction of predicted positives that are true positives.                                        | High precision reduces false alarms, avoiding unnecessary treatment.   | Reliable positive diagnosis measure.         | Can be high despite missing many true cases (low recall).                |
|                  | <b>Recall (Sensitivity, TPR)</b> | $\frac{TP}{TP + FN}$                                                                                          | Fraction of actual positives correctly identified.                                              | High recall reduces missed diagnoses; critical in pneumonia detection. | Captures ability to detect all cases.        | Often lowers precision (more false positives).                           |
|                  | <b>Specificity (TNR)</b>         | $\frac{TN}{TN + FP}$                                                                                          | Fraction of actual negatives correctly identified.                                              | High specificity reduces misclassification of healthy patients.        | Good at identifying normal cases.            | Gives no info on positive-case detection.                                |
|                  | <b>F1-Score</b>                  | $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$                       | Harmonic mean of precision and recall.                                                          | Balances false positives and false negatives in one score.             | Robust in imbalanced datasets.               | Less intuitive than precision/recall alone.                              |
|                  | <b>AUC</b>                       | $\int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt$                                                                  | Area under TPR vs. FPR curve over thresholds.                                                   | Measures threshold-independent discrimination ability.                 | Gold standard; robust to imbalance.          | Doesn't show performance at chosen clinical threshold.                   |
| Object Detection | <b>IoU</b>                       | $\frac{\text{Area}(B_{\text{pred}} \cap B_{\text{true}})}{\text{Area}(B_{\text{pred}} \cup B_{\text{true}})}$ | Overlap ratio between predicted and ground-truth boxes.                                         | Higher IoU = better localization.                                      | Direct spatial accuracy measure.             | Sensitive to small errors.                                               |
|                  | <b>mAP</b>                       | $\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i$<br>$\text{AP} = \int_0^1 p(r) dr$                         | Mean AP across classes/IoU thresholds; $\text{AP} = \text{area under precision-recall curve}$ . | Captures classification and localization across thresholds.            | Comprehensive performance metric.            | Less intuitive; sensitive to IoU choice.                                 |

**Table 5**

Comparative summary of datasets, model architectures, tasks, evaluation metrics, and key performance results for chest X-ray (CXR) analysis.

| Dataset                        | Model Architecture     | Specific Model/Technique           | Task                       | AUC          | F1-Score     | Accuracy     | Sensitivity  | mAP          | Reference                             |
|--------------------------------|------------------------|------------------------------------|----------------------------|--------------|--------------|--------------|--------------|--------------|---------------------------------------|
| <b>Kermany's CXR</b>           | Ensemble               | GoogLeNet, ResNet-18, DenseNet-121 | Classification (3-class)   | –            | <b>0.988</b> | <b>0.988</b> | <b>0.988</b> | –            | Kundu et al. (2021) [48]              |
|                                | Attention-Enhanced CNN | DenseNet-121 with CBAM             | Classification (3-class)   | <b>0.990</b> | <b>0.986</b> | <b>0.986</b> | <b>0.988</b> | –            | Rashid et al. (2025) [49]             |
| <b>NIH ChestX-ray14</b>        | Standard CNN           | CheXNet (DenseNet-121)             | Multi-label Classification | <b>0.768</b> | <b>0.435</b> | –            | –            | –            | Rajpurkar et al. (2017) [32]          |
|                                | Standard CNN           | Location-aware DenseNet-121        | Multi-label Classification | <b>0.731</b> | –            | –            | –            | –            | Guendel et al. (2018) [50]            |
| <b>RSNA Pneumonia CheXpert</b> | Object Detector        | Improved Faster R-CNN              | Lung Opacity Detection     | –            | –            | –            | –            | <b>0.392</b> | Xu j et al. (2023) [51]               |
|                                | Zero-Shot Learning     | MoCoCLIP (CLIP-based)              | Multi-label Classification | <b>0.750</b> | –            | –            | –            | –            | Bhardwaj P et al. (2025) [52]         |
|                                | Interpretable CNN      | ResNet50 with LRP                  | Multi-label Classification | <b>0.930</b> | –            | <b>0.910</b> | –            | –            | Colin, J., & Surantha, N. (2025) [53] |
| <b>MIMIC-CXR</b>               | Object Detector        | Fast-YOLO                          | Pathology Detection        | –            | –            | –            | <b>0.947</b> | <b>0.978</b> | Zhao B et al. (2025) [54]             |
| <b>VinDr-CXR</b>               | Federated Learning     | Federated CNN Model                | Multi-label Classification | <b>0.943</b> | –            | –            | –            | –            | Lotfinia M et al. (2025) [55]         |



**Fig. 2.** Performance comparison of techniques and datasets, showing key evaluation metrics for chest X-ray (CXR) analysis.

ray14, the initial CheXNet model set a more modest but realistic benchmark with an F1-score of 0.435, a number that subsequent models have found difficult to significantly improve through architectural changes alone. This underscores the significant impact of data quality and task complexity on reported results. However, as will be shown, even these verified norms may be fundamentally flawed. The critical paper by Zech et al. (2018) [38] demonstrates that a model with a high internal test AUC of 0.931 can see its performance decline to 0.815 on an external dataset, not because it failed to learn, but because it learned non-clinical, dataset-specific shortcuts. This analytical context must be kept in mind when evaluating the current state-of-the-art performance.

## 4. Discussion

### 4.1. Principal findings and prevailing trends

This comprehensive review synthesizes the extensive research on DL for pneumonia detection, highlighting a field characterized by rapid technological advancements but facing significant obstacles to practical implementation. Our analysis of 73 studies reveals three main outcomes.

First, the database is fundamentally fractured. As discussed in Table 2, there is a clear trade-off between the size of huge, weakly labeled datasets like NIH ChestX-ray14 and the high-fidelity annotations of smaller, curated collections like Kermany's CXR. This division considerably influences model performance and the comparability of

results across studies, making a single "leaderboard" way of assessing models particularly problematic.

Second, the architectural environment is dominated by mature and data-efficient CNNs. Our technical assessment in Table 5 shows that while new architectures like Vision Transformers show promise, the field's workhorses remain fundamental CNNs like ResNet and DenseNet. Their high inductive biases make them particularly useful for image tasks, especially in a domain where vast, perfectly labeled datasets are the exception, not the rule. Hybrid CNN-Transformer models provide the most promising research horizon, trying to mix the best of both paradigms.

Third, and most critically, performance on static benchmarks is a poor and sometimes misleading proxy for therapeutic usefulness. Our evaluation of assessment measures in Table 5, along with the XAI analysis in Table 6, indicates that high scores in accuracy or AUC could hide underlying model defects. The frequent issue of shortcut learning has exposed the fragility of models trained in one environment, forcing a fundamental modification in how we evaluate and trust these systems.

### 4.2. The "Shortcut Learning" Pitfall and the Crisis of generalization

The most fundamental challenge revealed by this investigation is that models struggle to generalize to out-of-distribution data. The Zech et al. (2018) study exemplifies this, where a model with strong internal validity failed when tested at an external institution. The model had not

**Table 6**

Summary of datasets, model architectures, specific techniques, tasks, evaluation metrics, and performance outcomes for chest X-ray (CXR) analysis.

| Dataset                                           | Model / Approach   | XAI Technique Category               | Specific XAI Method                      | XAI Impact on Performance  | Primary XAI Contribution                                    | Reference                                                 |
|---------------------------------------------------|--------------------|--------------------------------------|------------------------------------------|----------------------------|-------------------------------------------------------------|-----------------------------------------------------------|
| Kermamy's CXR                                     | D-ProtoPNet        | Prototypical / Case-Based (Ante-hoc) | Ante-hoc Prototypical Parts              | Performance-Negative       | Clinical Rationale (Intuitive explanations)                 | Yazdani, E., et al. (2025). [56]                          |
| Kermamy's CXR                                     | ResNet50           | Multiple Post-hoc                    | LRP, CAMs, Adversarial Training          | Performance-Positive (LRP) | Method Comparison (LRP improved performance)                | Colin, J., & Surantha, N. (2025). [53]                    |
| Cross-Institutional (MSH / NIH / IU)              | CNN (DenseNet-121) | Model Behavior Analysis (Auditing)   | Hospital-System Sorting, Activation Maps | Invalidated Performance    | Shortcut Detection (Model learned hospital artifacts)       | Zech, J. R., et al. (2018). [38]                          |
| Aggregated public datasets (incl. NIH ChestX-ray) | CNN                | Model Behavior Analysis (Auditing)   | Saliency Maps, GANs                      | Invalidated Performance    | Shortcut Detection (Model learned confounding text markers) | DeGrave, A. J., Janizek, J. D., & Lee, S. I. (2021). [57] |

actually learned the radiological signs of pneumonia but instead learned to identify the hospital of origin as a key confounding variable. This is not an isolated case; similar shortcut learning has been seen in models that associated sickness with text marks or patient positioning common in portable radiography.

This “shortcut learning” phenomenon is not merely a technical curiosity; it poses a major patient safety hazard. A model that focuses on hospital-specific artifacts can consistently misdiagnose patients from an underrepresented institution, leading to delayed treatment or wasteful operations. This reality significantly weakens the “train once, deploy anywhere” paradigm and needs a new focus on domain generalization and resilience as major research goals.

#### 4.3. The unmet need for subtype classification

A key goal of this research was to evaluate the use of deep learning (DL) for pneumonia subtype classification. Our findings reveal that this is a significantly under-researched area. While the Kermamy dataset provides the necessary labels, most studies focus on the binary classification of normal versus pneumonia. Differentiating etiologies has important clinical implications for treatment decisions. The lack of emphasis on this area indicates a substantial gap between current research efforts and clinical needs.

#### 5. Limitations

This review, however exhaustive, is subject to considerable limits. First, the analysis is constrained by the quality and clarity of the underlying material. As our risk of bias assessment indicates, many studies fail to give appropriate clarification on their approach, hindering replication. The variability in evaluation procedures and the inconsistent reporting of performance indicators across studies a problem we addressed by providing standardized tables made direct model-to-model comparison unfeasible and demonstrates a systemic issue in the field. Second, there is a well-known publication bias towards excellent results. Finally, our research is constrained to models evaluated on publicly available datasets, omitting studies done on vast, proprietary institutional datasets, which may contain significant biases and create different conclusions.

#### 6. Future research directions

Based on the outcomes of this investigation, a clear roadmap for improving the science towards clinically robust AI may be given. The focus must change from obtaining incremental success on existing benchmarks to tackling the core challenges of generalization and trustworthiness.

Development of Robust Datasets: Future efforts should focus on quality and diversity. This includes producing additional huge, expertly annotated datasets with localized labels (e.g., bounding boxes or semantic segmentations) and curating multi-institutional datasets

developed particularly to test out-of-distribution generalization.

Focus on Clinically Relevant Tasks: A bigger focus should be on clinically relevant but under-studied areas, such as pneumonia subtype classification.

Necessary XAI-Based Validation: XAI should be regarded as a routine, necessary phase in the validation pipeline, utilized not only for interpretability but also as a debugging tool to actively probe for shortcut learning and verify models are depending on clinically important characteristics.

Prospective Clinical Trials: Ultimately, the value of any diagnostic AI must be confirmed in the clinical arena. The profession must move beyond retrospective research and do prospective, multi-center, randomized controlled studies that investigate the influence of these tools on clinical workflow, diagnostic accuracy, and, most importantly, patient outcomes.

#### 7. Conclusion

This comprehensive study aimed to answer five key questions about the state of deep learning for pneumonia diagnosis. Our review of 73 studies offers clear, evidence-based answers. We found that the research landscape is built on a foundation of split data (RQ1), with a significant trade-off between the availability of weakly labeled datasets and the quality of smaller, expertly annotated ones. The field is largely dominated by data-efficient Convolutional Neural Networks, with hybrid models representing the most promising area (RQ2). Notably, while models show expert-level performance on certain benchmarks (RQ3), our analysis reveals that these metrics can be misleading as the sole indicators of clinical relevance. The common problem of “shortcut learning,” where models exploit dataset-specific artifacts instead of learning true pathology, undermines many high-performance claims and is the biggest barrier to generalization (RQ5). Additionally, we identified a significant gap between current research and clinical needs, with essential tasks like pneumonia subtype characterization being poorly addressed (RQ4). The main conclusion of this study is that, although the technical potential of deep learning for pneumonia diagnosis is clear, its application for robust, widespread clinical use is severely limited by issues of generality, reproducibility, and validation. Moving forward, research must shift its focus from pursuing top results on flawed standards to developing trustworthy, generalizable, and clinically validated systems.

#### CRedit authorship contribution statement

**Rupali Vyas:** Writing – original draft, Resources, Methodology.  
**Rajendra Kumar Pandey:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial



interests or personal relationships that could have appeared to influence the work reported in this paper.

## References

- [1] L. Macpherson, et al., Risk factors for death among children aged 5–14 years hospitalised with pneumonia: a retrospective cohort study in Kenya, *BMJ Glob. Health* 4 (5) (2019) e001715, <https://doi.org/10.1136/bmjgh-2019-001715>.
- [2] A. Torres, et al., Pneumonia, *Nat. Rev. Dis. Primers* 7 (1) (2021) 25, <https://doi.org/10.1038/s41572-021-00259-0>.
- [3] C. Lin, B. Ye, C. Ping, L. Li, Pneumonia review based on pathogenic factors, *Curr. Traditional Med.* 10 (2023) 1–12, <https://doi.org/10.2174/0122150838255055230919074231>.
- [4] F.T. Porras, C. Rodriguez, D. Rodriguez, P. Lezama, R. Inquilla, Y. Pomachagua. Deep Learning Algorithms in Chest Images for Pneumonia Detection, in: *Proceedings - 2022 14th IEEE International Conference on Computational Intelligence and Communication Networks, CICN 2022*, pp. 442–447, 2022, doi: 10.1109/CICN56167.2022.10008321.
- [5] I.F. Jassam, S.M. Elkaffas, A.A. El-Zoghbi, Chest X-ray pneumonia detection by dense-net, in: *2021 31st International Conference on Computer Theory and Applications (ICCTA)*, IEEE, 2021, pp. 176–179, <https://doi.org/10.1109/ICCTA54562.2021.9916637>.
- [6] S. Anand, R. K. Roshan, and D. S. M. Chest X ray image enhancement using deep contrast diffusion learning. *Optik (Stuttg.)*. 279, p. 170751. 2023, doi: 10.1016/J.IJLEO.2023.170751.
- [7] E. Çalli, E. Sogancioglu, B. van Ginneken, K.G. van Leeuwen, K. Murphy, Deep learning for chest X-ray analysis: a survey, *Med. Image Anal.* 72 (2021) 102125, <https://doi.org/10.1016/J.MEDIA.2021.102125>.
- [8] A.K. Jaiswal, P. Tiwari, S. Kumar, D. Gupta, A. Khanna, J.J.P.C. Rodrigues, Identifying pneumonia in chest X-rays: a deep learning approach, *Measurement (Lond)*. 145 (2019) 511–518, <https://doi.org/10.1016/j.measurement.2019.05.076>.
- [9] M.S. Al Reshan, et al., Detection of pneumonia from chest X-ray images utilizing MobileNet model, *Healthcare* 11 (11) (2023) 1561, <https://doi.org/10.3390/HEALTHCARE11111561>.
- [10] M. Li, Y. Jiang, Y. Zhang, H. Zhu, Medical image analysis using deep learning algorithms, *Front. Public Health*. 11 (2023), <https://doi.org/10.3389/fpubh.2023.1273253>.
- [11] M. Puttagunta, S. Ravi, Medical image analysis based on deep learning approach, *Multimed. Tools Appl.* 80 (16) (2021) 24365–24398, <https://doi.org/10.1007/S11042-021-10707-4>.
- [12] X. Chen, et al., Recent advances and clinical applications of deep learning in medical image analysis, *Med. Image Anal.* 79 (2022) 102444, <https://doi.org/10.1016/j.media.2022.102444>.
- [13] S. Atasever, N.U.H. Azginoglu, D.S. Terzi, R. Terzi, A comprehensive survey of deep learning research on medical image analysis with focus on transfer learning, *Clin. Imaging* 94 (2023) 18–41, <https://doi.org/10.1016/J.CLINIMAG.2022.11.003>.
- [14] P.K. Wong, et al., Automatic detection of multiple types of pneumonia: open dataset and a multi-scale attention network, *Biomed. Signal Process Control*. 73 (2022), <https://doi.org/10.1016/J.BSPC.2021.103415>.
- [15] A. Kareem, H. Liu, P. Sant, Review on pneumonia image detection: a machine learning approach, *Hum.-Centric Intel. Syst.* 2 (1) (2022) 31–43, <https://doi.org/10.1007/S44230-022-00002-2>.
- [16] W. Khan, N. Zaki, L. Ali, Intelligent pneumonia identification from chest X-rays: a systematic literature review, *IEEE Access* 9 (2021) 51747–51771, <https://doi.org/10.1109/ACCESS.2021.3069937>.
- [17] D.J. Alapat, M.V. Menon, S. Ashok, A review on detection of pneumonia in chest X-ray images using neural networks, *J. Biomed. Phys. Eng.* 12 (6) (2022) 551, <https://doi.org/10.31661/JBPE.V010.2202-1461>.
- [18] S. Sharma, K. Guleria, A systematic literature review on deep learning approaches for pneumonia detection using chest X-ray images, *Multimed. Tools Appl.* 83 (8) (2024) 24101–24151, <https://doi.org/10.1007/S11042-023-16419-1/METRICS>.
- [19] P.F. Whiting, et al., QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies, *Ann. Intern. Med.* 155 (8) (2011) 529–536, <https://doi.org/10.7326/0003-4819-155-8-201110180-00009>.
- [20] D.S. Kermany, et al., Identifying medical diagnoses and treatable diseases by image-based deep learning, *Cell* 172 (5) (2018) 1122–1131.e9, <https://doi.org/10.1016/j.cell.2018.02.010>.
- [21] G. Shih, et al., Augmenting the national institutes of health chest radiograph dataset with expert annotations of possible pneumonia, *Radiol. Artif. Intell.* 1 (1) (2019), <https://doi.org/10.1148/RXAI.2019180041/ASSET/IMAGES/LARGE/RXAI.2019180041.FIG3.JPEG>.
- [22] H.Q. Nguyen, et al., VinDr-CXR: an open dataset of chest X-rays with radiologist's annotations, *Sci. Data* 9 (1) (2022), <https://doi.org/10.1038/s41597-022-01498-w>.
- [23] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers. ChestX-ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. 2017. [Online]. Available: <https://uts.nlm.nih.gov/metathesaurus.html>.
- [24] "CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison." Accessed: Apr. 30, 2025. [Online]. Available: <https://stanfordmlgroup.github.io/competitions/chexpert/>.
- [25] A.E.W. Johnson, et al., MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports, *Sci. Data* 6 (1) (2019) 317, <https://doi.org/10.1038/s41597-019-0322-0>.
- [26] D. Kikoo, B. Tamin, S. Hardjadilaga, I.A. Iswanto, Using various convolutional neural network to detect pneumonia from chest X-ray images: a systematic literature review, *JOIV: Int. J. Informatics Visualization* 7 (2) (2023) 310, <https://doi.org/10.30630/joiv.7.2.1015>.
- [27] M. M. Abubakar, B. Z. Adamu, and M. Z. Abubakar. Pneumonia Classification Using Hybrid CNN Architecture. in: *2021 International Conference on Data Analytics for Business and Industry, ICDABI 2021*, pp. 520–522, 2021, doi: 10.1109/ICDABI53623.2021.9655918.
- [28] D. Varshni, K. Thakral, L. Agarwal, R. Nijhawan, A. Mittal, Pneumonia detection using CNN based feature extraction, in: *Proceedings of 2019 3rd IEEE International Conference on Electrical, Computer and Communication Technologies, ICECCT 2019*, 2019, <https://doi.org/10.1109/ICECCT.2019.8869364>.
- [29] J. Deng, W. Dong, R. Socher, L.J. Li, K. Li, L. Fei-Fei, ImageNet: A large-scale hierarchical image database, in: *2009 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, 2009*, pp. 248–255, <https://doi.org/10.1109/CVPR.2009.5206848>.
- [30] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. 2016-December, pp. 770–778. 2016, doi: 10.1109/CVPR.2016.90.
- [31] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger. Densely connected convolutional networks. *Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017*. 2017, pp. 2261–2269. 2017, doi: 10.1109/CVPR.2017.243.
- [32] P. Rajpurkar et al. CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning. Nov. 2017, [Online]. Available: <https://arxiv.org/abs/1711.05225v3>.
- [33] N. Habib, M.M. Hasan, M.M. Reza, M.M. Rahman, Ensemble of CheXNet and VGG-19 feature extractor with random forest classifier for pediatric pneumonia detection, *SN Comput. Sci.* 1 (6) (2020) 1–9, <https://doi.org/10.1007/S42979-020-00373-X/FIGURES/10>.
- [34] R. Vyas, D.R. Khadkar, Ensemble of deep learning architectures with machine learning for pneumonia classification using chest X-rays, *J. Imaging Informatics Med.* 2024 (2024) 1–20, <https://doi.org/10.1007/S10278-024-01201-Y>.
- [35] M.R. Hasan, S.M. Azmat Ullah, S. Md Rabiul Islam, Recent advancement of deep learning techniques for pneumonia prediction from chest X-ray image, *Med. Rep.* 7 (2024) 100106, <https://doi.org/10.1016/j.hmedic.2024.100106>.
- [36] A. Bustos, A. Pertusa, J.-M. Salinas, M. de la Iglesia-Vayá, PadChest: a large chest x-ray image dataset with multi-label annotated reports, *Med. Image Anal.* 66 (2020) 101797, <https://doi.org/10.1016/j.media.2020.101797>.
- [37] D. Demner-Fushman, et al., Preparing a collection of radiology examinations for distribution and retrieval, *J. Am. Med. Inform. Assoc.* 23 (2) (2016) 304–310, <https://doi.org/10.1093/jamia/ocv080>.
- [38] J.R. Zech, M.A. Badgeley, M. Liu, A.B. Costa, J.J. Titano, E.K. Oermann, Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study, *PLoS Med.* 15 (11) (2018) e1002683, <https://doi.org/10.1371/journal.pmed.1002683>.
- [39] S. Ren, K. He, R. Girshick, and J. Sun. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks." [Online]. Available: <https://github.com/>.
- [40] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: unified, real-time object detection, in: *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2016, pp. 779–788, <https://doi.org/10.1109/CVPR.2016.91>.
- [41] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollar, Focal loss for dense object detection, in: *2017 IEEE International Conference on Computer Vision (ICCV)*, IEEE, 2017, pp. 2999–3007, <https://doi.org/10.1109/ICCV.2017.324>.
- [42] S. Sharma, K. Guleria, A systematic literature review on deep learning approaches for pneumonia detection using chest X-ray images, *Multimed. Tools Appl.* 83 (8) (2023) 24101–24151, <https://doi.org/10.1007/s11042-023-16419-1>.
- [43] A. Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. Jun. 2021, [Online]. Available: <http://arxiv.org/abs/2010.11929>.
- [44] K. Simonyan and A. Zisserman. Very Deep Convolutional Networks for Large-Scale Image Recognition. Apr. 2015, [Online]. Available: <http://arxiv.org/abs/1409.1556>.
- [45] C. Szegedy, et al., Going deeper with convolutions, in: *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2015, pp. 1–9, <https://doi.org/10.1109/CVPR.2015.7298594>.
- [46] A. G. Howard et al. MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. 2017, [Online]. Available: <http://arxiv.org/abs/1704.04861>.
- [47] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the Inception Architecture for Computer Vision. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. 2016, pp. 2818–2826. 2015, doi: 10.1109/CVPR.2016.308.
- [48] R. Kundu, R. Das, Z.W. Geem, G.-T. Han, R. Sarkar, Pneumonia detection in chest X-ray images using an ensemble of deep learning models, *PLoS One* 16 (9) (2021) e0256630, <https://doi.org/10.1371/journal.pone.0256630>.
- [49] A.B. Rashid, J. Asma, K. Barua, D. Das, An enhanced deep learning framework for pneumonia detection in chest X-rays, *SN Comput. Sci.* 6 (5) (May 2025) 472, <https://doi.org/10.1007/s42979-025-04017-x>.
- [50] S. Guendel et al. Learning to recognize Abnormalities in Chest X-Rays with Location-Aware Dense Networks. 2018, [Online]. Available: <http://arxiv.org/abs/1803.04565>.

- [51] J. Xu, H. Ren, S. Cai, X. Zhang, An improved faster R-CNN algorithm for assisted detection of lung nodules, *Comput. Biol. Med.* 153 (2023) 106470, <https://doi.org/10.1016/j.combiomed.2022.106470>.
- [52] P. Bhardwaj, S. Bhat, and A. Maier. Enhancing zero-shot learning in medical imaging: integrating clip with advanced techniques for improved chest x-ray analysis. Mar. 2025, [Online]. Available: <http://arxiv.org/abs/2503.13134>.
- [53] J. Colin, N. Surantha, Interpretable deep learning for pneumonia detection using chest X-ray images, *Information* 16 (1) (2025) 53, <https://doi.org/10.3390/info16010053>.
- [54] B. Zhao, L. Chang, Z. Liu, Fast-YOLO network model for X-ray image detection of pneumonia, *Electronics (Basel)*. 14 (5) (2025) 903, <https://doi.org/10.3390/electronics14050903>.
- [55] M. Lotfinia, A. Tayebiarasteh, S. Samiei, M. Joodaki, S. Tayebi Arasteh, Boosting multi-demographic federated learning for chest radiograph analysis using general-purpose self-supervised representations, *Eur. J. Radiol. Artif. Intell.* 3 (2025) 100028, <https://doi.org/10.1016/j.ejrai.2025.100028>.
- [56] E. Yazdani, A. Neizehbaz, N. Karamzade-Ziarati, S.R. Kheradpisheh, Explainable artificial intelligence for pneumonia classification: clinical insights into deformable prototypical part network in pediatric chest x-ray images, *J. Med. Imaging Radiat. Sci.* 56 (5) (2025) 102023, <https://doi.org/10.1016/j.jmir.2025.102023>.
- [57] A.J. DeGrave, J.D. Janizek, S.I. Lee, AI for radiographic COVID-19 detection selects shortcuts over signal, *Nat. Mach. Intell.* 3 (7) (2021) 610–619, <https://doi.org/10.1038/s42256-021-00338-7>.